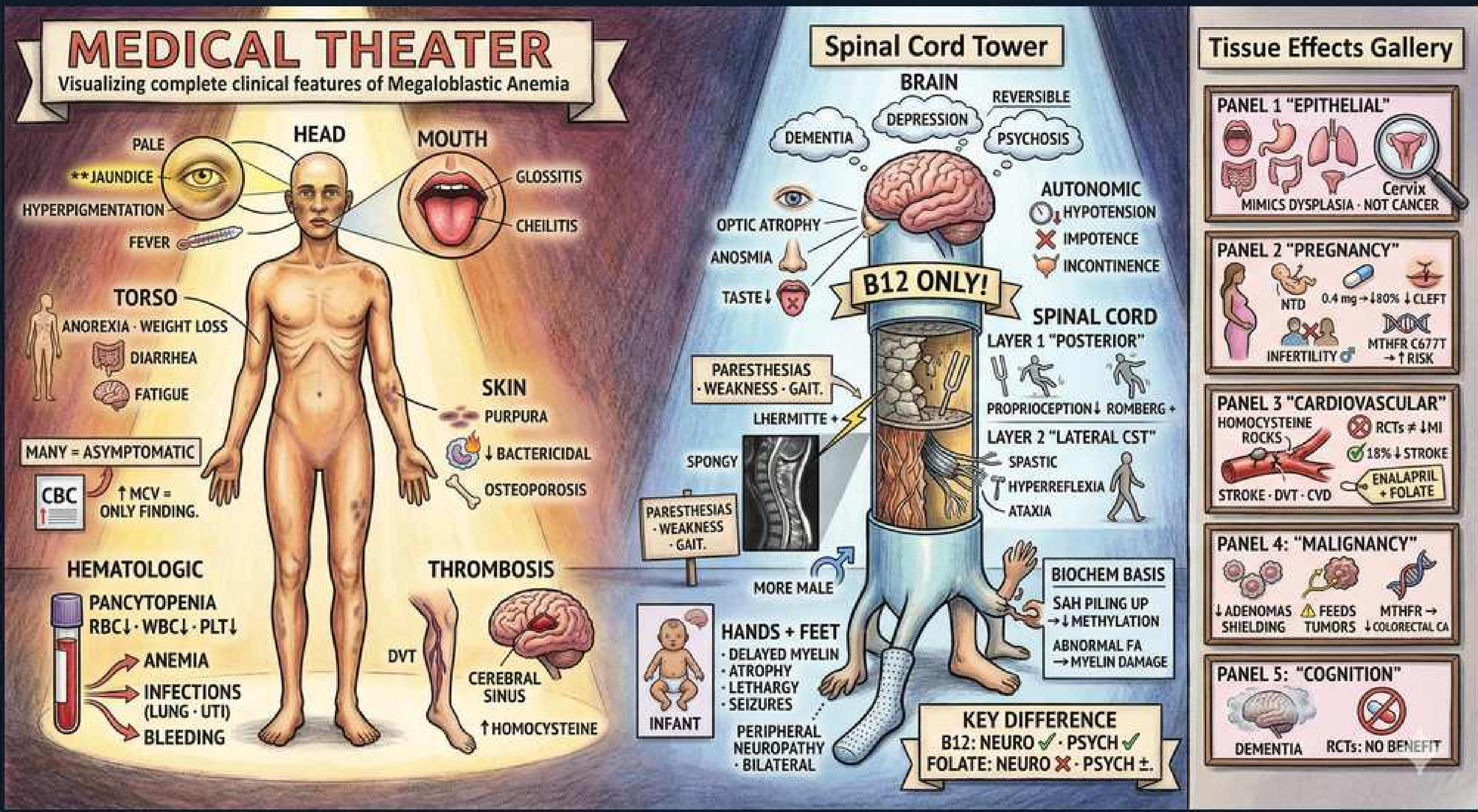
Megaloblastic — B12 Clinical Features & Neurology



1. Pallor & lemon-yellow jaundice

WHAT YOU SEE

HEAD 'PALE', 'JAUNDICE', 'HYPERPIGMENTATION'

WHAT IT ENCODES

Pallor reflects anemia; the classic LEMON-YELLOW tint blends pallor with mild jaundice from intramedullary hemolysis (ineffective erythropoiesis destroys megaloblasts in the marrow before they ever leave). This raises indirect (unconjugated) bilirubin and LDH while haptoglobin falls — a hemolysis-without-bleeding pattern. Skin/mucosal HYPERPIGMENTATION is a less-known but characteristic B12 finding that reverses with replacement.

BOARD TRAP

WRONG: choosing autoimmune hemolytic anemia because of jaundice + low haptoglobin. Discriminator: in megaloblastic anemia the Coombs (DAT) is NEGATIVE, MCV is HIGH (>100), reticulocytes are LOW (production failure), and the smear shows oval macrocytes + hypersegmented neutrophils — not spherocytes with a brisk reticulocytosis.

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2. Glossitis & cheilitis

WHAT YOU SEE

MOUTH 'GLOSSITIS', 'CHEILITIS'

WHAT IT ENCODES

Rapidly dividing epithelium is hit first by impaired DNA synthesis. The tongue becomes smooth, beefy-red, and sore (atrophic glossitis = Hunter glossitis), and the mouth corners crack (angular cheilitis/stomatitis). Mnemonic: B12/folate hit the FAST cells — gut lining, mouth, marrow. These reverse within days-weeks of replacement.

BOARD TRAP

WRONG: attributing a smooth red tongue + cheilitis only to iron deficiency. Both iron and B12/folate cause glossitis and angular cheilitis; the discriminator is the MCV — LOW/microcytic points to iron, HIGH/macrocytic points to B12 or folate. A combined deficiency can give a normal MCV with a high RDW (dimorphic).

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3. Pancytopenia

WHAT YOU SEE

HEMATOLOGIC 'PANCYTOPENIA RBC↓ WBC↓ PLT↓'

WHAT IT ENCODES

Because B12/folate are needed for DNA synthesis in ALL marrow lineages, severe deficiency causes pancytopenia (anemia + leukopenia + thrombocytopenia), not isolated anemia. The marrow is HYPERcellular (ineffective production), distinguishing it from aplastic anemia (hypocellular). Mnemonic: ineffective hematopoiesis = full factory, empty trucks.

BOARD TRAP

WRONG: jumping to bone-marrow biopsy and calling pancytopenia + a blast-like marrow an acute leukemia or MDS. Discriminator: check B12, folate, MMA, and homocysteine FIRST — megaloblastic pancytopenia corrects completely with vitamin replacement, whereas leukemia does not. Treating reversible deficiency avoids unnecessary chemotherapy.

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4. Often asymptomatic — high MCV

WHAT YOU SEE

CBC tube '↑MCV ONLY FINDING', 'MANY = ASYMPTOMATIC'

WHAT IT ENCODES

Deficiency develops slowly (B12 stores last 3-4 years), so many patients compensate and are asymptomatic; an incidentally HIGH MCV (often >100, sometimes >110-115 fL) may be the only abnormality. The earliest morphologic clue is hypersegmented neutrophils, which can precede a frank rise in MCV. Mnemonic: MCV up, patient fine — still work it up.

BOARD TRAP

WRONG: ignoring an isolated macrocytosis as 'asymptomatic = benign.' Discriminator: an unexplained high MCV mandates a B12/folate workup, and a normal hemoglobin does NOT exclude early or neurologic B12 deficiency.

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5. Thrombosis (homocysteine)

WHAT YOU SEE

THROMBOSIS 'DVT', 'CEREBRAL VEIN', '↑HOMOCYSTEINE'

WHAT IT ENCODES

B12 (and folate) are cofactors that recycle homocysteine to methionine via methionine synthase; deficiency raises homocysteine, which is prothrombotic and damages endothelium. This increases venous (DVT, cerebral venous sinus) and arterial thrombosis risk. Mnemonic: high homocysteine = sticky blood + sick vessels.

BOARD TRAP

WRONG: working up a young patient's unprovoked DVT/cerebral vein thrombosis only for inherited thrombophilias while missing hyperhomocysteinemia. Discriminator: check homocysteine (and B12/folate) — it is elevated in BOTH B12 and folate deficiency, so it flags the deficiency but does not distinguish which one.

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6. B12 ONLY — neuro

WHAT YOU SEE

Central spinal cord tower 'B12 ONLY!'

WHAT IT ENCODES

Neurologic disease is the cardinal feature that separates B12 from folate deficiency: B12 deficiency causes subacute combined degeneration (SCD) of the cord plus peripheral neuropathy and cognitive change, whereas folate deficiency does NOT cause neuro signs. The mechanism is failure of methionine synthase → low methionine/SAM → impaired myelin methylation (and adenosyl-B12 failure → toxic methylmalonyl-CoA). Mnemonic: B12 = Brain + 12 = the vitamin with neurology.

BOARD TRAP

WRONG (the classic board distractor): 'folate deficiency causes the same neurologic signs.' It does NOT — folate corrects the anemia but not the neurologic disease. This is why you must rule out B12 before giving folate alone.

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7. Posterior column — proprioception

WHAT YOU SEE

'LAYER 1 POSTERIOR', 'PROPRIOCEPTION', 'ROMBERG'

WHAT IT ENCODES

SCD degenerates the dorsal (posterior) columns first → loss of vibration sense and proprioception, sensory ataxia, and a POSITIVE Romberg sign (falls when eyes close because visual compensation is removed). Findings are symmetric and start distally in the legs. Mnemonic: posterior columns carry Position + Vibration → Romberg positive.

BOARD TRAP

WRONG: calling a wide-based gait with a positive Romberg 'cerebellar ataxia.' Discriminator: cerebellar ataxia is present with eyes OPEN and does NOT worsen with eye closure (Romberg negative); dorsal-column/sensory ataxia is Romberg POSITIVE. Check B12/MMA before chasing a cerebellar lesion.

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8. Lateral CST — spastic

WHAT YOU SEE

'LAYER 2 LATERAL CST', 'SPASTIC', 'HYPERREFLEXIA', 'ATAXIA'

WHAT IT ENCODES

SCD also degenerates the lateral corticospinal tracts → upper-motor-neuron signs: spasticity, hyperreflexia, and extensor (upgoing) plantar/Babinski. The combination of UMN signs WITH absent ankle reflexes (coexisting peripheral neuropathy) is a classic mixed picture. Mnemonic: 'combined' = dorsal columns + lateral CST degenerating together.

BOARD TRAP

WRONG: a paraparesis with hyperreflexia and upgoing toes prompting an MRI-only chase for cord compression while ignoring B12. Discriminator: SCD is typically symmetric, subacute, and reversible early; always send B12 + MMA. Untreated, SCD becomes IRREVERSIBLE — early treatment is the whole point.

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9. Paresthesias hands & feet

WHAT YOU SEE

'HANDS + FEET', 'PERIPHERAL NEUROPATHY BILATERAL'

WHAT IT ENCODES

The earliest and most common neuro complaint is symmetric distal paresthesias (numbness/tingling) in a stocking-glove (hands AND feet) pattern from peripheral neuropathy. This can precede anemia and is the most reversible component. Mnemonic: tingling toes and fingertips = check B12.

BOARD TRAP

WRONG: attributing bilateral stocking-glove paresthesias automatically to diabetic neuropathy in a diabetic patient. Discriminator: many diabetics are on METFORMIN, which lowers B12 absorption — always check a B12 level (and MMA if borderline) before blaming diabetes alone.

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10. Dementia / psych / psychosis

WHAT YOU SEE

BRAIN 'DEMENTIA', 'DEPRESSION', 'PSYCHOSIS' (reversible)

WHAT IT ENCODES

B12 deficiency causes cognitive impairment, depression, irritability, and frank psychosis ('megaloblastic madness') — a REVERSIBLE cause of dementia. Mechanism: impaired methylation/myelin maintenance in the brain. Mnemonic: B12 is on every reversible-dementia checklist (along with thyroid, neurosyphilis, NPH).

BOARD TRAP

WRONG: diagnosing a confused older patient with Alzheimer dementia without screening B12. Discriminator: B12 deficiency is a TREATABLE/reversible dementia — a low B12 (especially with high MMA) demands replacement; failure to check it misses a curable cause.

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11. Neuro can precede anemia

WHAT YOU SEE

KEY FACT 'B12: NEURO + PSYCH', timeline

WHAT IT ENCODES

B12 neurologic disease can appear BEFORE any anemia and even with a NORMAL MCV (~25% of neuro cases lack macrocytosis or anemia). Therefore CBC is not a screen for neurologic B12 deficiency — treat on clinical suspicion with B12 + MMA/homocysteine. Mnemonic: nerves can fail while the blood still looks normal.

BOARD TRAP

WRONG: 'a normal CBC/normal MCV excludes B12 neuro disease.' It does NOT — send B12 and confirm with MMA + homocysteine (both elevated → B12). Waiting for anemia risks irreversible cord damage.

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12. Folate: neuro absent

WHAT YOU SEE

PANEL 5 COGNITION, folate 'NO BENEFIT', neuro X

WHAT IT ENCODES

Folate corrects the megaloblastic ANEMIA but provides NO benefit to neurologic disease — the key B12-vs-folate discriminator. Worse, giving folate to a B12-deficient patient can normalize the blood count while the dorsal-column/CST degeneration silently progresses. Mnemonic: folate fixes the marrow, B12 fixes the nerves.

BOARD TRAP

WRONG (high-yield): treating any macrocytic anemia with folate empirically. Discriminator: ALWAYS exclude/replace B12 first; folate monotherapy in occult B12 deficiency masks the anemia and can precipitate or worsen subacute combined degeneration.
